

MIGRATING TO A TEMPORAL SCHEMA

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PG DATA 2026

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OUTLINE

- Temporal Tables
- Queries
- DML
- DDL

GOALS

-
- ☒ Data Preserving
 - ☒ Backwards Compatible
 - ☒ Incremental Progress
-
- ☐ Online Migration

APPLICATION TIME

clients				
id	name	price	valid_from	valid_til
1	SpaceZ	\$130	Jan 2018	Jan 2019
1	SpaceZ	\$140	Jan 2019	
2	Ifily	\$120	Jan 2016	

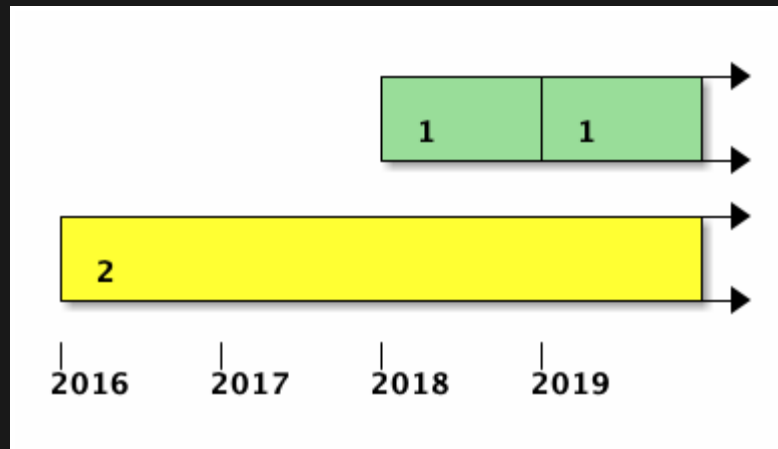
APPLICATION TIME

clients			
id	name	rate	valid_at
1	SpaceZ	\$130	[Jan 2018, Jan 2019)
1	SpeceZ	\$140	[Jan 2019,)
2	Ifily	\$120	[Jan 2016,)

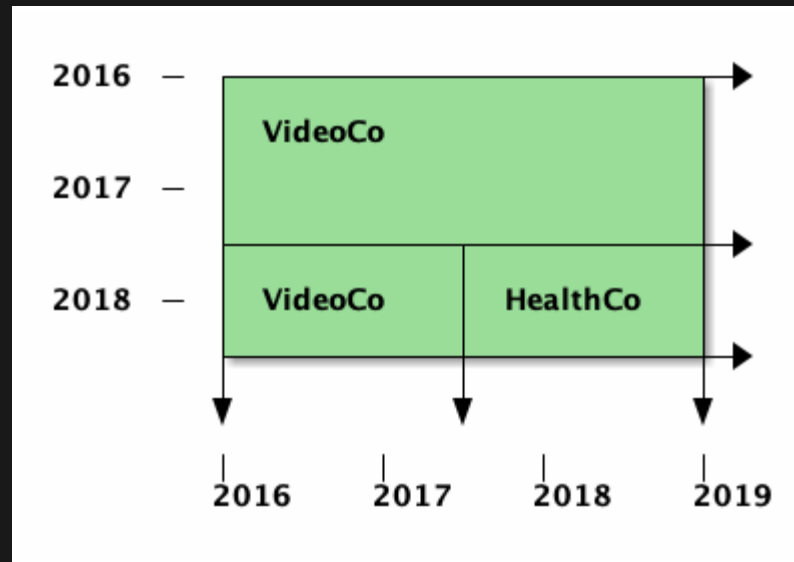
APPLICATION TIME

clients			
id	name	rate	valid_at
1 1	SpaceZ SpeceZ	\$130 \$140	[Jan 2018, Jan 2019) [Jan 2019,)
2	Ifily	\$120	[Jan 2016,)
3 3	Nadir Nadir	\$100 \$110	[Jan 2016,) [Jan 2017, Jan 2018)

APPLICATION TIME



SYSTEM TIME



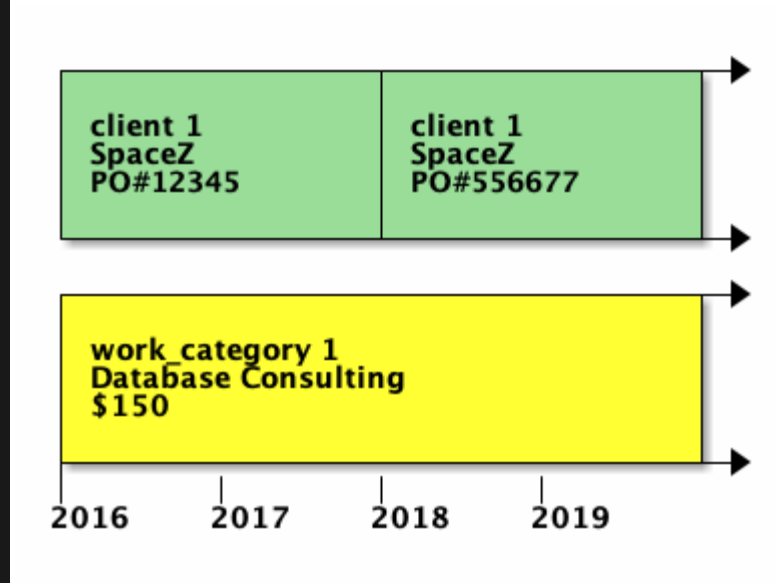
PRIMARY KEYS

```
-- old:  
CONSTRAINT legacy_pk EXCLUDE  
  (id WITH =, valid_at WITH &&)  
  
-- new:  
CONSTRAINT shiny_pk PRIMARY KEY  
  (id, valid_at WITHOUT OVERLAPS)
```

FOREIGN KEYS

```
CONSTRAINT fk_work_categories_on_client_id  
  FOREIGN KEY (client_id, PEROID valid_at)  
  REFERENCES clients (id, PERIOD valid_at)
```

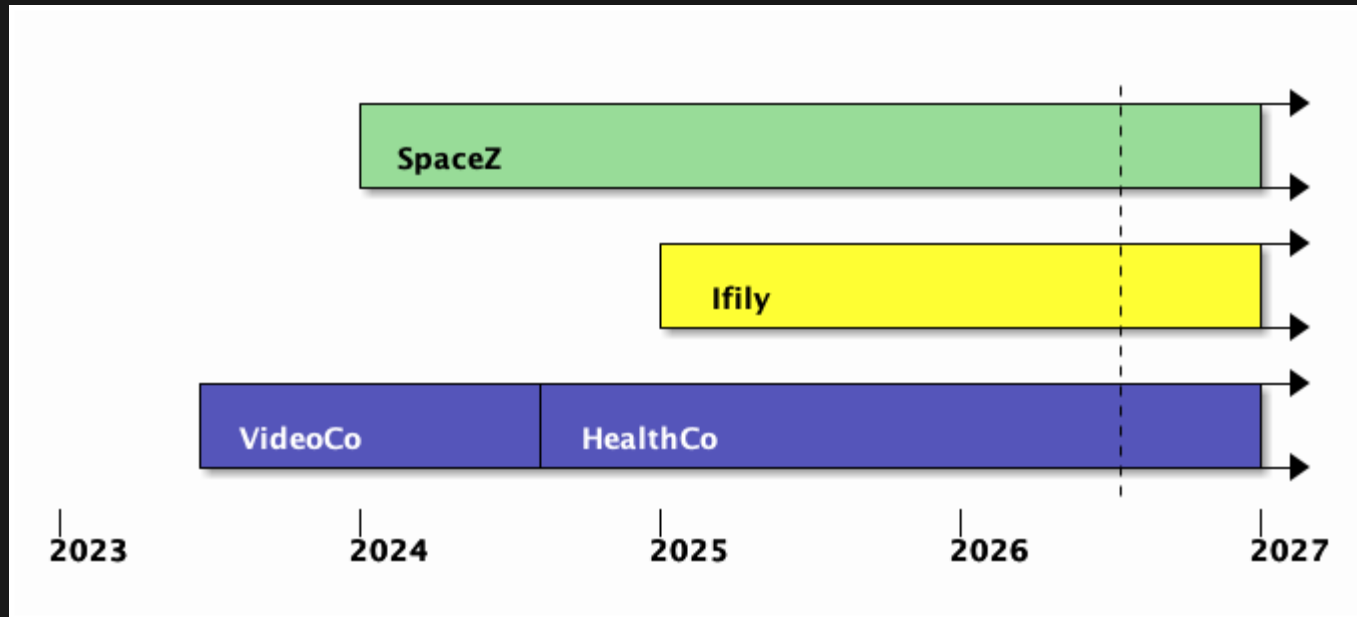
FOREIGN KEYS



FOREIGN KEYS

referencing	referenced	supported?
normal	normal	☒
temporal	temporal	☒
temporal	normal	☒
normal	temporal	☐

QUERIES



QUERIES

```
SELECT * EXCEPT valid_at  
FROM t  
WHERE valid_at @> current_timestamp;
```

QUERIES

$$\sigma(R \bowtie S) = \sigma(R) \times \sigma(S)$$

We now examine whether the valid-time algebra is in some sense a consistent extension of the snapshot algebra. An algebra is said to *reduce* to the snapshot algebra if taking a snapshot of the result of applying a valid-time operator on one or two valid-time relations is identical to the result of applying the analogous snapshot operator to the snapshots (at the same times) of the valid-time relation(s). Because the temporal algebra allows tuples that contain attributes of differing timestamps, it satisfies this property only through the introduction of distinguished nulls when taking snapshots. We avoid this problem by proving a weaker property: we restrict reducibility to operations on valid-time relations that have identical timestamps for all of their attributes, termed *homogeneous relations* [Gad88a].

Theorem 4 *The valid-time operators $\hat{\cup}$, $\hat{-}$, $\hat{\times}$, $\hat{\sigma}$, and $\hat{\pi}$ reduce to their snapshot counterparts when their arguments are homogeneous.*

VIEWS

```
1 SELECT id, name, slug,
2        ..., position,
3        -- created_at: The very first version's lower(valid_at):
4        -- range_agg does the right thing for NULL endpoints, unlike M
5        (SELECT lower(range_agg(valid_at))
6         FROM temporal.clients c2
7         WHERE c2.id = clients.id) AS created_at,
8        lower(valid_at) AS updated_at,
9        -- deleted_at: The very last version's upper(valid_at), which
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16 WHERE valid_at @> current_timestamp at time zone 'UTC'
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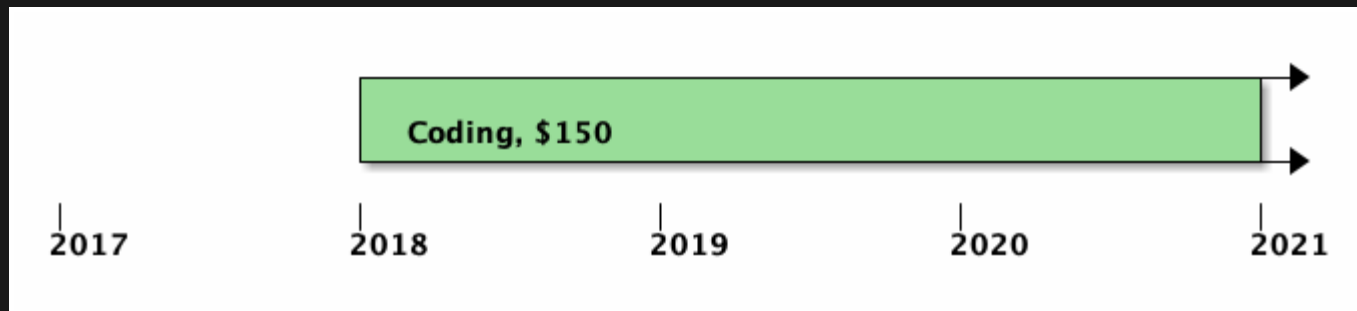
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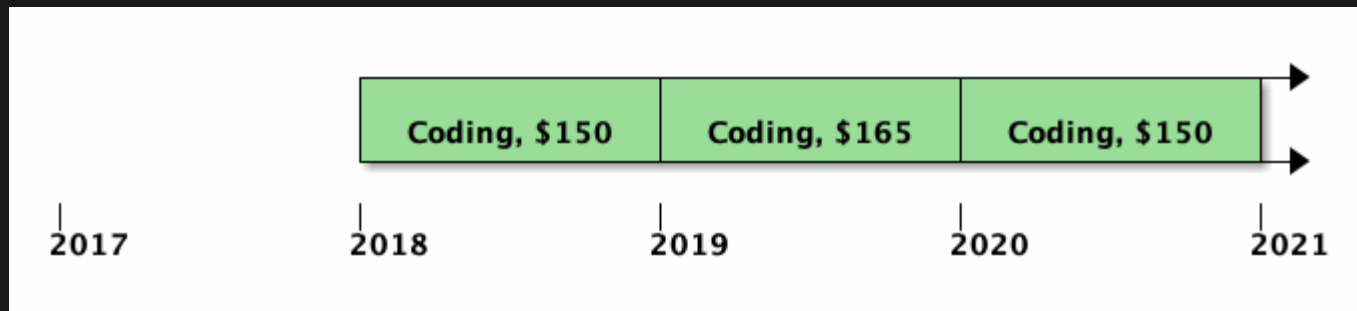
UPDATE

```
UPDATE work_categories  
  FOR PORTION OF valid_at  
  FROM '2019-01-01' TO '2020-01-01'  
  SET rate = 1.10 * rate;  
  WHERE client_id = 5;
```

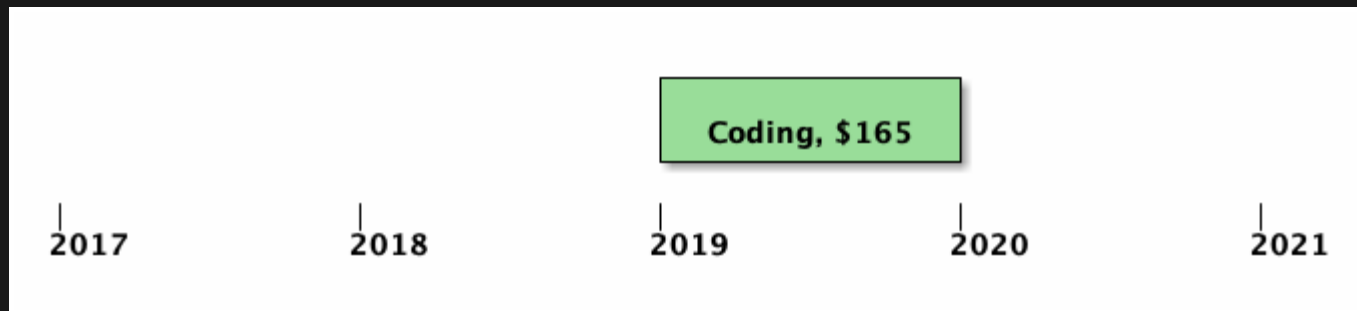

UPDATE



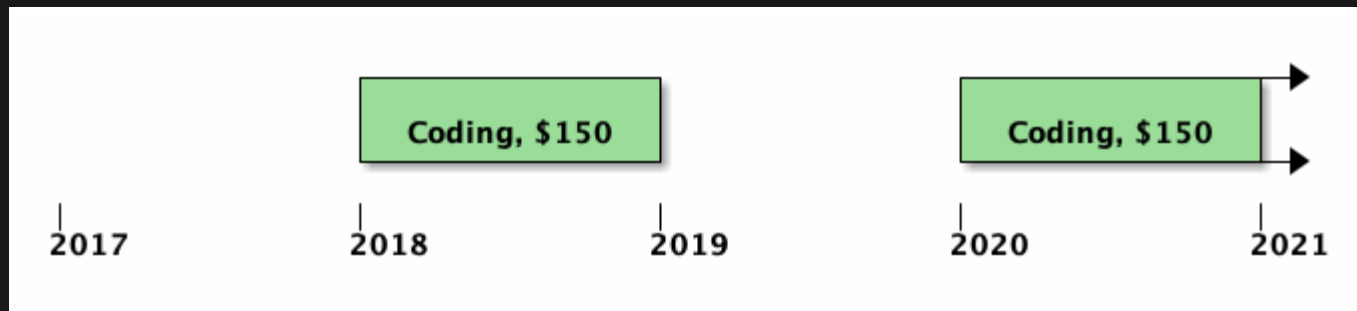
UPDATE



UPDATE



UPDATE



UPDATE

```
1 CREATE FUNCTION update_work_categories_view() RETURNS trigger
2 LANGUAGE plpgsql AS
3 $$
4 BEGIN
5     UPDATE temporal.work_categories
6         FOR PORTION OF valid_at
7         FROM current_timestamp AT TIME ZONE 'UTC'
8         TO NULL
9         SET client_id = NEW.client_id,
10            name       = NEW.name,
11            rate       = NEW.rate
12     WHERE id = NEW.id;
13     RETURN NEW; -- Enables RETURNING
14 END;
15 $$;
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10        name      = NEW.name,
11        rate      = NEW.rate
12     WHERE id = NEW.id;
13     RETURN NEW; -- Enables RETURNING
14 END;
15 $$;
```

UPDATE

```
CREATE TRIGGER trg_update_work_categories  
  INSTEAD OF UPDATE ON work_categories  
  FOR EACH ROW  
  EXECUTE FUNCTION update_work_categories_view();
```


DELETE

```
DELETE FROM clients
  FOR PORTION OF valid_at
    FROM current_timestamp AT TIME ZONE 'UTC'
    TO NULL
  WHERE name = 'Nadir';
```

INSERT

```
1 CREATE FUNCTION insert_work_categories_view()  
2 RETURNS trigger  
3 LANGUAGE plpgsql  
4 AS  
5 $$  
6 BEGIN  
7     INSERT INTO temporal.work_categories  
8         (client_id, name, rate, valid_at) VALUES  
9         (NEW.client_id, NEW.name, NEW.rate,  
10        tsrange(current_timestamp AT TIME ZONE 'UTC', NULL))  
11        RETURNING id INTO NEW.id;  
12    RETURN NEW;  
13 END;  
14 $$;
```

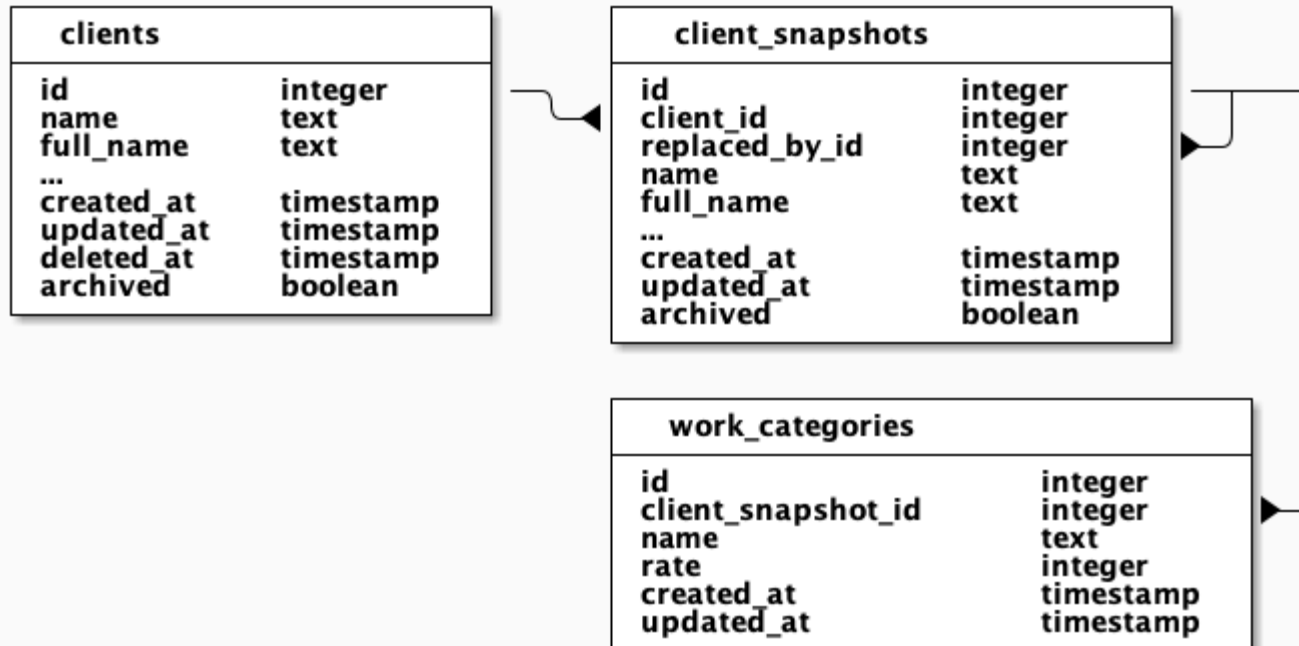
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12    RETURN NEW;  
13 END;  
14 $$;
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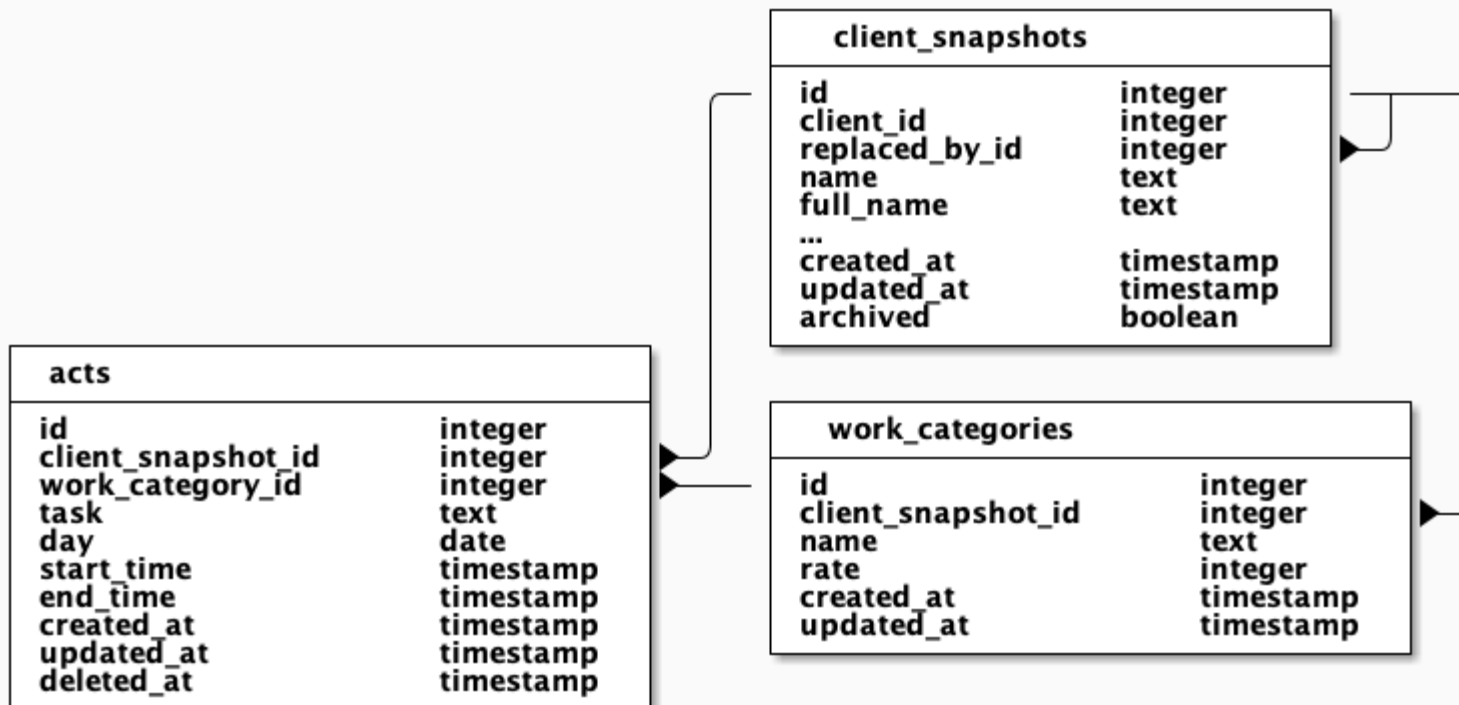
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10        tsrange(current_timestamp AT TIME ZONE 'UTC', NULL))  
11     RETURNING id INTO NEW.id;  
12     RETURN NEW;  
13 END;  
14 $$;
```

LEGACY TABLES



LEGACY TABLES



MIGRATION PLAN

- Clean up bad data.
 - Add more constraints.
- Create temporal schema with tables.
- Move data over from the old tables.
- Staging:
 - temporal.clients with client_snapshots view
 - remove client_snapshots
 - temporal.work_categories with view
 - remove work_categories view

INCREMENTAL PROGRESS

1. Add a dummy `valid_at` to the non-temporal table.
2. Put a traditionally-unique column on the temporal table.
3. Define a "loose" foreign key function.
4. Create separate id tables.

LOOSE FK

```
1 CREATE OR REPLACE FUNCTION add_loose_foreign_key(  
2   from_table regclass,  
3   fk_column  text,  
4   to_table   regclass,  
5   pk_column  text DEFAULT 'id',  
6   name       text DEFAULT NULL  
7 ) RETURNS void LANGUAGE plpgsql AS $fn$  
8 DECLARE  
9   v_suffix text := loose_fk_trigger_suffix(from_table, fk_column);  
10  v_ref_name text := 'loose_fk_ref_' || v_suffix;  
11  v_inv_name text := 'loose_fk_inv_' || v_suffix;  
12 BEGIN  
13   EXECUTE format(  
14     'CREATE CONSTRAINT TRIGGER %1$I  
15     AFTER INSERT OR UPDATE OF %2$I ON %3$I
```

LOOSE FK

```
10 v_ref_name text := 'loose_fk_ref_' || v_suffix;  
11 v_inv_name text := 'loose_fk_inv_' || v_suffix;  
12 BEGIN  
13     EXECUTE format(  
14         'CREATE CONSTRAINT TRIGGER %1$I  
15         AFTER INSERT OR UPDATE OF %2$I ON %3$s  
16         FROM %4$s  
17         NOT DEFERRABLE INITIALLY IMMEDIATE  
18         FOR EACH ROW  
19         EXECUTE PROCEDURE loose_fk_check_ref(%5$L, %6$L, %7$L,  
20         v_ref_name, fk_column, from_table, to_table,  
21         fk_column, to_table::text, pk_column  
22     );  
23  
24     EXECUTE format(  
25         'CREATE CONSTRAINT TRIGGER %1$I
```

LOOSE FK

```
19      EXECUTE PROCEDURE loose_fk_check_ref(%5$L, %6$L, %7$L,
20      v_ref_name, fk_column, from_table, to_table,
21      fk_column, to_table::text, pk_column
22  );
23
24  EXECUTE format(
25      'CREATE CONSTRAINT TRIGGER %1$I
26      AFTER UPDATE OF %2$I OR DELETE ON %3$s
27      FROM %4$s
28      NOT DEFERRABLE INITIALLY IMMEDIATE
29      FOR EACH ROW
30      EXECUTE PROCEDURE loose_fk_check_inv(%5$L, %6$L, %7$L,
31      v_inv_name, pk_column, to_table, from_table,
32      from_table::text, fk_column, pk_column
33  );
```

LOOSE FK

```
1 CREATE OR REPLACE FUNCTION loose_fk_check_ref()
2 RETURNS trigger LANGUAGE plpgsql AS $fn$
3 DECLARE
4     v_fk_column text := TG_ARGV[0];
5     v_to_table  text := TG_ARGV[1];
6     v_pk_column text := TG_ARGV[2];
7     v_is_null   boolean;
8     v_found     integer;
9     v_fk_value  text;
10 BEGIN
11     -- MATCH SIMPLE: a NULL fk passes the constraint.
12     EXECUTE format('SELECT ($1).%I IS NULL', v_fk_column)
13         USING NEW INTO v_is_null;
14     IF v_is_null THEN RETURN NULL; END IF;
15
```

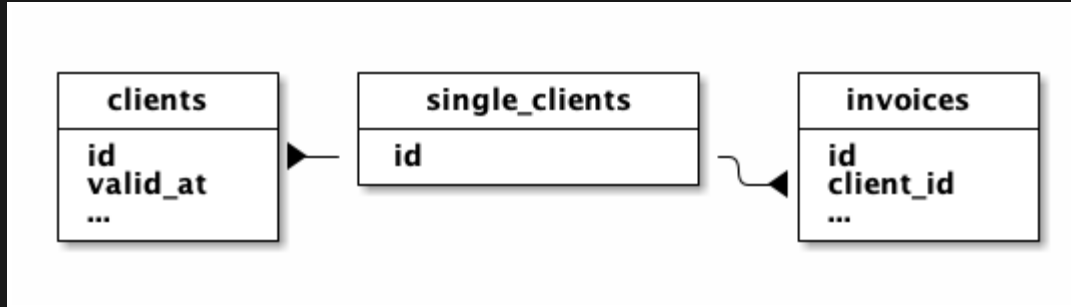
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5 v_fk_column text := TG_ARGV[1];
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10 BEGIN
11     -- MATCH SIMPLE: a NULL fk passes the constraint.
12     EXECUTE format('SELECT ($1).%I IS NULL', v_fk_column)
13         USING NEW INTO v_is_null;
14     IF v_is_null THEN RETURN NULL; END IF;
15
16     -- FOR KEY SHARE matches built-in RI: lock the referenced
17     -- concurrent UPDATE-of-key or DELETE blocks until we con
18     EXECUTE format(
19         'SELECT 1 FROM %1$s t
20         WHERE t.%2$I = ($1).%2$I FOR KEY SHARE OF t'
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LOOSE FK

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12 EXECUTE format('SELECT ($1).%I IS NULL', v_fk_column)
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18 EXECUTE format(
19     'SELECT 1 FROM %1$s t
20     WHERE t.%2$I = ($1).%3$I FOR KEY SHARE OF t',
21     v_to_table, v_pk_column, v_fk_column
22 ) USING NEW INTO v_found;
23
24 IF v_found IS NULL THEN
25     v_fk_value := to_jsonb(NEW) ->> v_fk_column;
26     RAISE foreign_key_violation USING
```

ID TABLES



```
CREATE TABLE temporal.single_clients (id integer PRIMARY KEY);
```

```
CREATE FUNCTION trg_generate_temporal_client_id()
```

```
RETURNS trigger LANGUAGE plpgsql
```

```
AS $$
```

```
BEGIN
```

```
IF NEW.id IS NULL THEN
```

```
    NEW.id := nextval(
```

```
        pg_get_serial_sequence('temporal.clients', 'id'));
```

```
END IF;
```

```
INSERT INTO temporal.single_clients (id) VALUES (NEW.id)
```

```
    ON CONFLICT DO NOTHING;
```

```
RETURN NEW;
```

END;

\$\$;

CLIENTS

```
1 CREATE TABLE temporal.clients (  
2   id integer NOT NULL GENERATED BY DEFAULT AS IDENTITY  
3   /* REFERENCES temporal.single_clients (id) */,  
4   valid_at tsrange NOT NULL,  
5   ...,  
6   position integer NOT NULL DEFAULT 0,  
7   archived boolean NOT NULL DEFAULT false,  
8   client_snapshot_id SERIAL NOT NULL,  
9   CONSTRAINT clients_pkey PRIMARY KEY (id, valid_at WITHOUT OVERLAPS),  
10  CONSTRAINT clients_slug_key UNIQUE (slug, valid_at WITHOUT OVERLAPS)  
11  -- acts_as_list will trip this if we check  
12  -- in the middle of a transaction (same as for non-temporal):  
13  CONSTRAINT clients_position_key UNIQUE (position, valid_at WITHOUT O  
14  DEFERRABLE INITIALLY DEFERRED,  
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17 EOF
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```
1 CREATE TABLE temporal.clients (  
2   id integer NOT NULL GENERATED BY DEFAULT AS IDENTITY  
3   /* REFERENCES temporal.single_clients (id) */,  
4   valid_at tsrange NOT NULL,  
5   ...,  
6   position integer NOT NULL DEFAULT 0,  
7   archived boolean NOT NULL DEFAULT false,  
8   client_snapshot_id SERIAL NOT NULL,  
9   CONSTRAINT clients_pkey PRIMARY KEY (id, valid_at WITHOUT OVERLAPS),  
10  CONSTRAINT clients_slug_key UNIQUE (slug, valid_at WITHOUT OVERLAPS)  
11  -- acts_as_list will trip this if we check  
12  -- in the middle of a transaction (same as for non-temporal):  
13  CONSTRAINT clients_position_key UNIQUE (position, valid_at WITHOUT O  
14  DEFERRABLE INITIALLY DEFERRED,  
15  CONSTRAINT clients_client_snapshot_id_key UNIQUE (client_snapshot_id  
16 );  
17 EOF
```

CLIENTS

```
CREATE OR REPLACE FUNCTION trg_generate_temporal_client_snapsh
RETURNS trigger
LANGUAGE plpgsql
AS
$$
BEGIN
    SELECT nextval(
        pg_get_serial_sequence('temporal.clients',
                                'client_snapshot_id'))
        INTO NEW.client_snapshot_id;
    RETURN NEW;
END;
$$;
```

CLIENTS

```
1 INSERT INTO temporal.clients
2 (id, ..., valid_at, name, position, client_snapshot_id)
3 SELECT cs.client_id, cs.name, ...,
4        tsrange(
5            cs.created_at,
6            lead(cs.created_at)
7              OVER (PARTITION BY cs.client_id ORDER BY cs.created_at)),
8        dense_rank()
9          OVER (ORDER BY c.position
10              ROWS UNBOUNDED PRECEDING EXCLUDE TIES)
11          AS position,
12        cs.id
13 FROM client_snapshots cs
14 JOIN clients c ON c.id = cs.client_id;
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13 FROM client_snapshots cs
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```

client_snapshots

```
SELECT client_snapshot_id AS id,  
       id AS client_id,  
       lead(client_snapshot_id)  
         OVER (PARTITION BY id ORDER BY id)  
         AS replaced_by_id,  
       name,  
       ...,  
       lower(valid_at) AS created_at,  
       -- In practice the old data often has created_at <> up  
       -- because we set replaced_by_id,  
       -- but the app doesn't care if we lie about it:  
       lower(valid_at) AS updated_at,  
FROM temporal.clients
```

work_categories

```
SELECT original_id AS id,  
       c.client_snapshot_id,  
       wc.name, wc.rate, wc.position,  
       -- created_at: The very first version's lower(valid_at)  
       -- range_agg does the right thing for NULL endpoints,  
       (SELECT lower(range_agg(valid_at))  
        FROM temporal.clients c2  
        WHERE c2.id = clients.id) AS created_at,  
       lower(wc.valid_at) AS updated_at,  
       wc.retainer_hours  
FROM   temporal.work_categories wc  
JOIN   temporal.clients c  
ON     c.id = wc.client_id AND c.valid_at && wc.valid_at  
WHERE  wc.valid_at @> current_timestamp at time zone 'UTC'
```

THANKS!

<https://github.com/pjungwir/migrating-to-a-temporal-schema>

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- Paul Jungwirth. These slides. <https://github.com/pjungwir/migrating-to-a-temporal-schema>